

Normal Distribution Function

In 1773 by De-Moivre, as a *limiting case of the binomial distribution*.

In 1809 by Gauss, as *distribution of errors in Astronomy*.

Definition: A r.v. X is said to have a normal distribution with parameters μ and σ^2 if its probability density function is given by:

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\left\{\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\}} ; \quad -\infty < x < \infty$$
$$-\infty < \mu < \infty$$
$$\sigma > 0$$

Mean μ is the location parameter and

Standard deviation σ is the scale parameter

The normal distribution was introduced by the French mathematician Abraham de Moivre in 1733 and was used by him to approximate probabilities associated with binomial random variables when the binomial parameter n - is large.

This result was later extended by Laplace and others and is now encompassed in a probability theorem known as the central limit theorem, which gives a theoretical base to the often noted empirical observation that, in practice, many random phenomena obey, at least approximately, a normal probability distribution.

Some examples of this behavior are

- the height of a person
- the velocity in any direction of a molecule in gas
- the error made in measuring a physical quantity

Central limit theorem

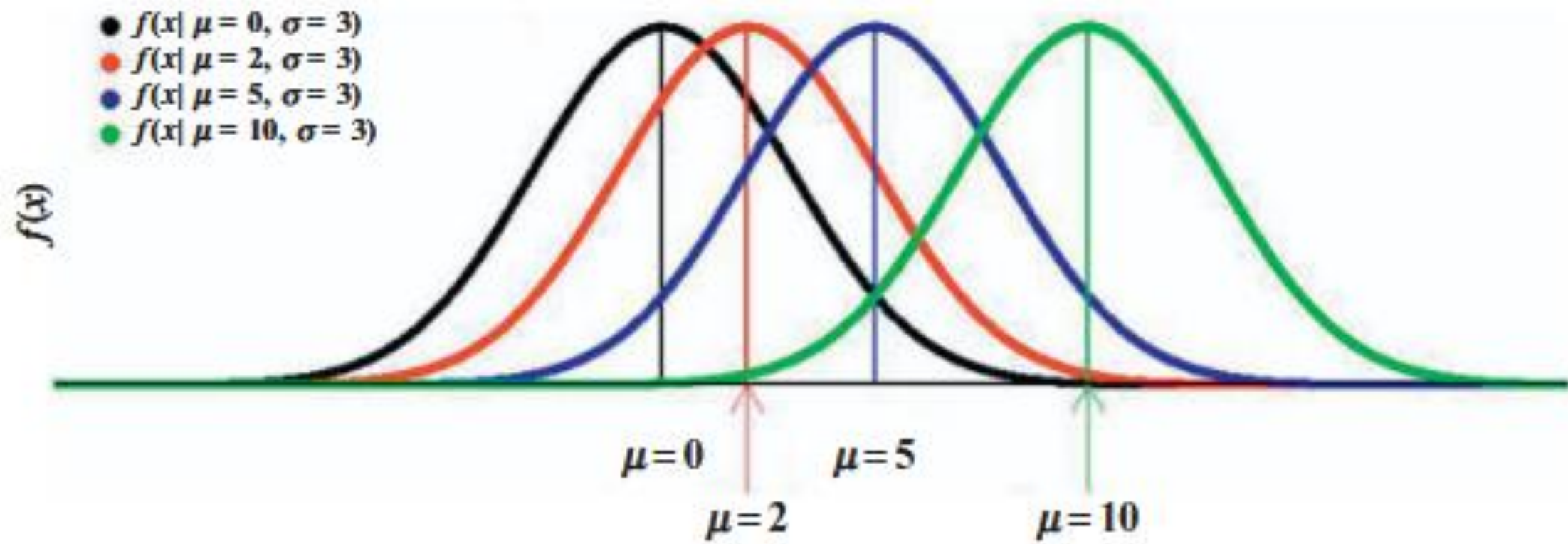
Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables, each having mean $E(X_i) = \mu$ and variance $V(X_i) = \sigma^2$, $i = 1, 2, \dots, n$

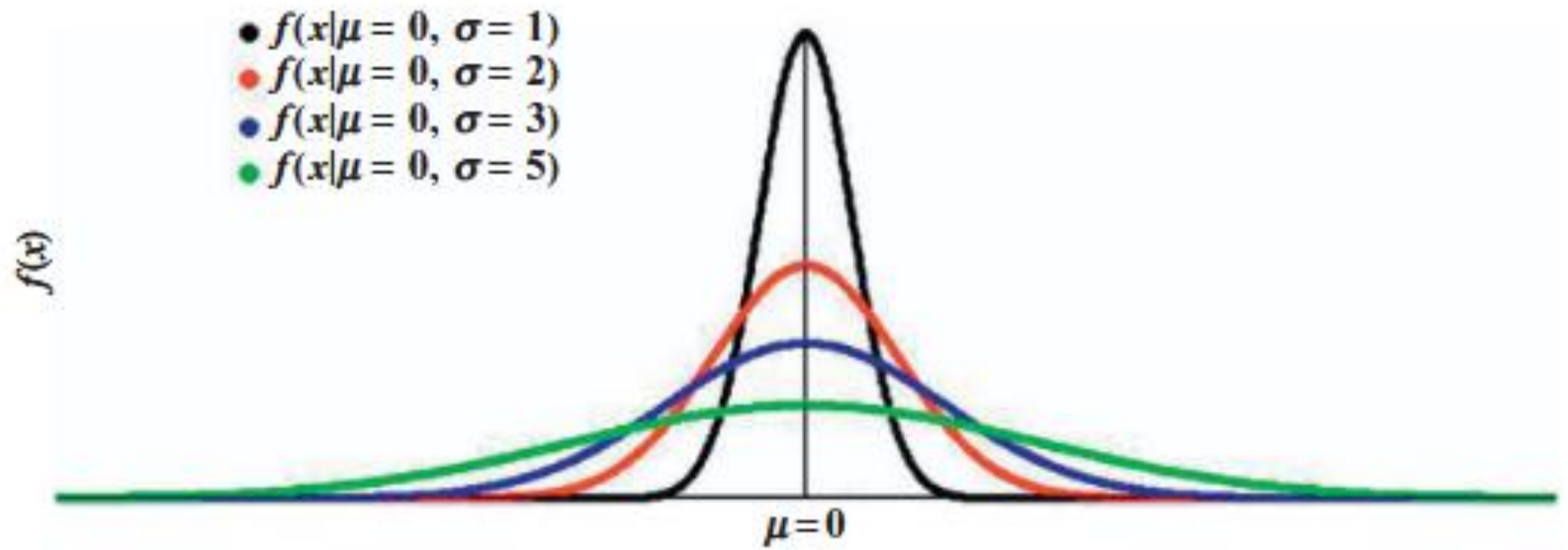
Then for n is large ($n \rightarrow \infty$), the distribution of $\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$

$$\frac{X_1 + X_2 + \dots + X_n - n\mu}{\sigma\sqrt{n}}$$

tends to standard normal with mean $n\mu$ and variance $n\sigma^2$. *i.e.*, $-\infty < a < \infty$

$$P\left\{\frac{X_1 + X_2 + \dots + X_n - n\mu}{\sigma\sqrt{n}} \leq a\right\} \rightarrow \frac{1}{\sqrt{2\pi}} \int_{-\infty}^a e^{-x^2/2} dx \quad \text{as } n \rightarrow \infty$$





Properties:

1. If $X \sim N(\mu, \sigma^2)$, then $\mathbf{Z} = \frac{X-\mu}{\sigma}$, is a standard normal variate with $E(Z) = 0$ and $V(Z) = 1$ and we write $\mathbf{Z} \sim N(\mathbf{0}, \mathbf{1})$.

The p.d.f. of the standard normal variate $\mathbf{Z}$ is given by:

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \quad ; \quad -\infty < z < \infty$$

and the corresponding distribution function is given by:

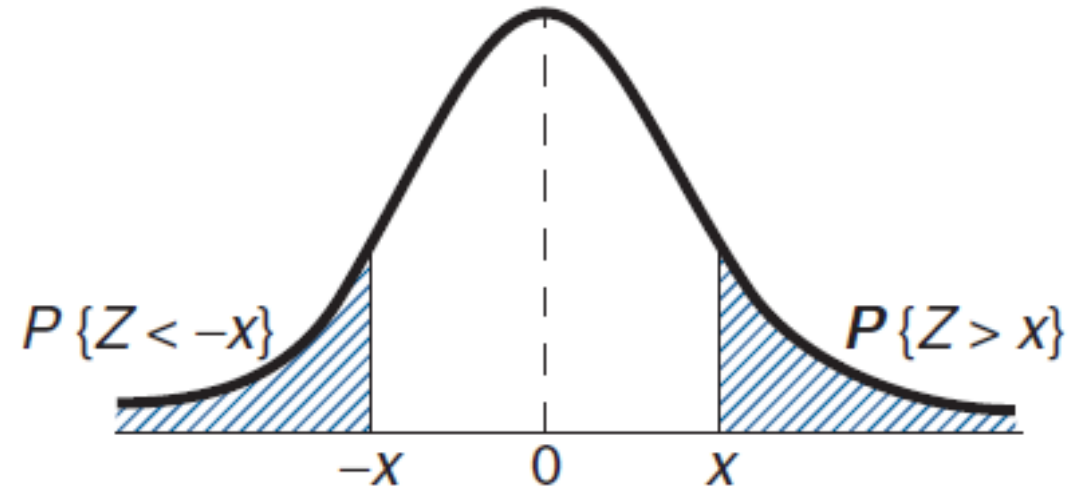
$$\Phi(z) = P(Z \leq z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du$$

$$\begin{aligned}
2. \quad P(a \leq X \leq b) &= P\left(\frac{a-\mu}{\sigma} \leq \frac{X-\mu}{\sigma} \leq \frac{b-\mu}{\sigma}\right) \\
&= P\left(\frac{a-\mu}{\sigma} \leq Z \leq \frac{b-\mu}{\sigma}\right) \\
&= P\left(Z \leq \frac{b-\mu}{\sigma}\right) - P\left(Z \leq \frac{a-\mu}{\sigma}\right) \\
P(a \leq X \leq b) &= \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)
\end{aligned}$$

$$\begin{aligned}
3. \quad P(X < b) &= P\left(\frac{X-\mu}{\sigma} < \frac{b-\mu}{\sigma}\right) \\
&= P\left(Z < \frac{b-\mu}{\sigma}\right) \\
&= \Phi\left(\frac{b-\mu}{\sigma}\right)
\end{aligned}$$

4. Symmetrical about mean.

$$\begin{aligned}\Phi(-x) &= P\{Z < -x\} \\ &= P\{Z > x\} \\ &= 1 - \Phi(x)\end{aligned}$$



Thus, for instance,

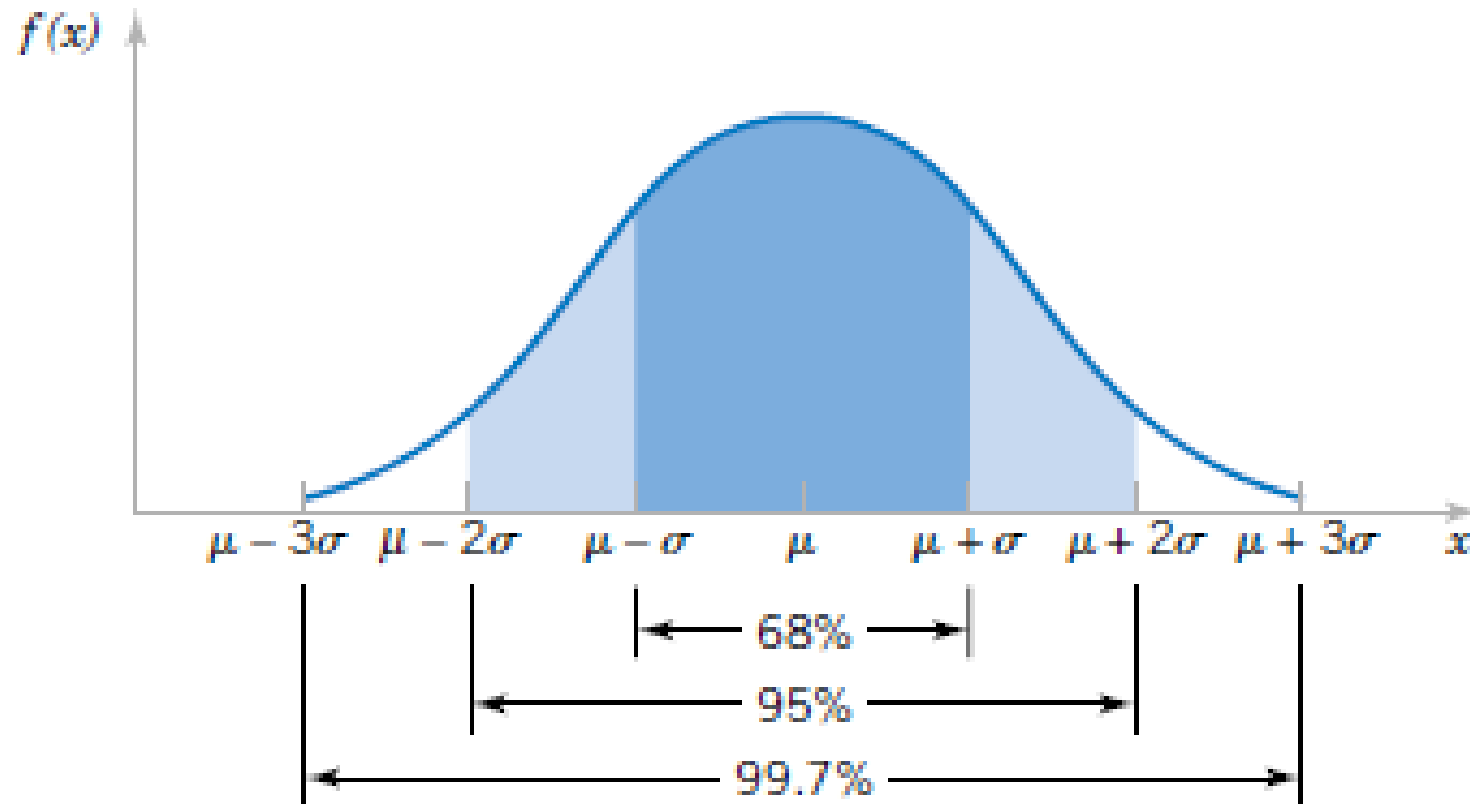
$$P\{Z < -1\} = \Phi(-1) = 1 - \Phi(1) = 1 - 0.8413 = 0.1587$$

For any normal random variables,

$$P(\mu - \sigma < X < \mu + \sigma) = P(-1 < Z < 1) = 0.6827$$

$$P(\mu - 2\sigma < X < \mu + 2\sigma) = P(-2 < Z < 2) = 0.9543$$

$$P(\mu - 3\sigma < X < \mu + 3\sigma) = P(-3 < Z < 3) = 0.9973$$



Moments

Mean

$$\mu$$

Variance

$$\sigma^2$$

Coefficient of Variation

$$\sigma/\mu$$

Median

$$\mu$$

Mean Deviation

$$\sqrt{\frac{2\sigma^2}{\pi}}$$

Coefficient of Skewness

$$0$$

Coefficient of Kurtosis

$$3$$

Moment generating Function

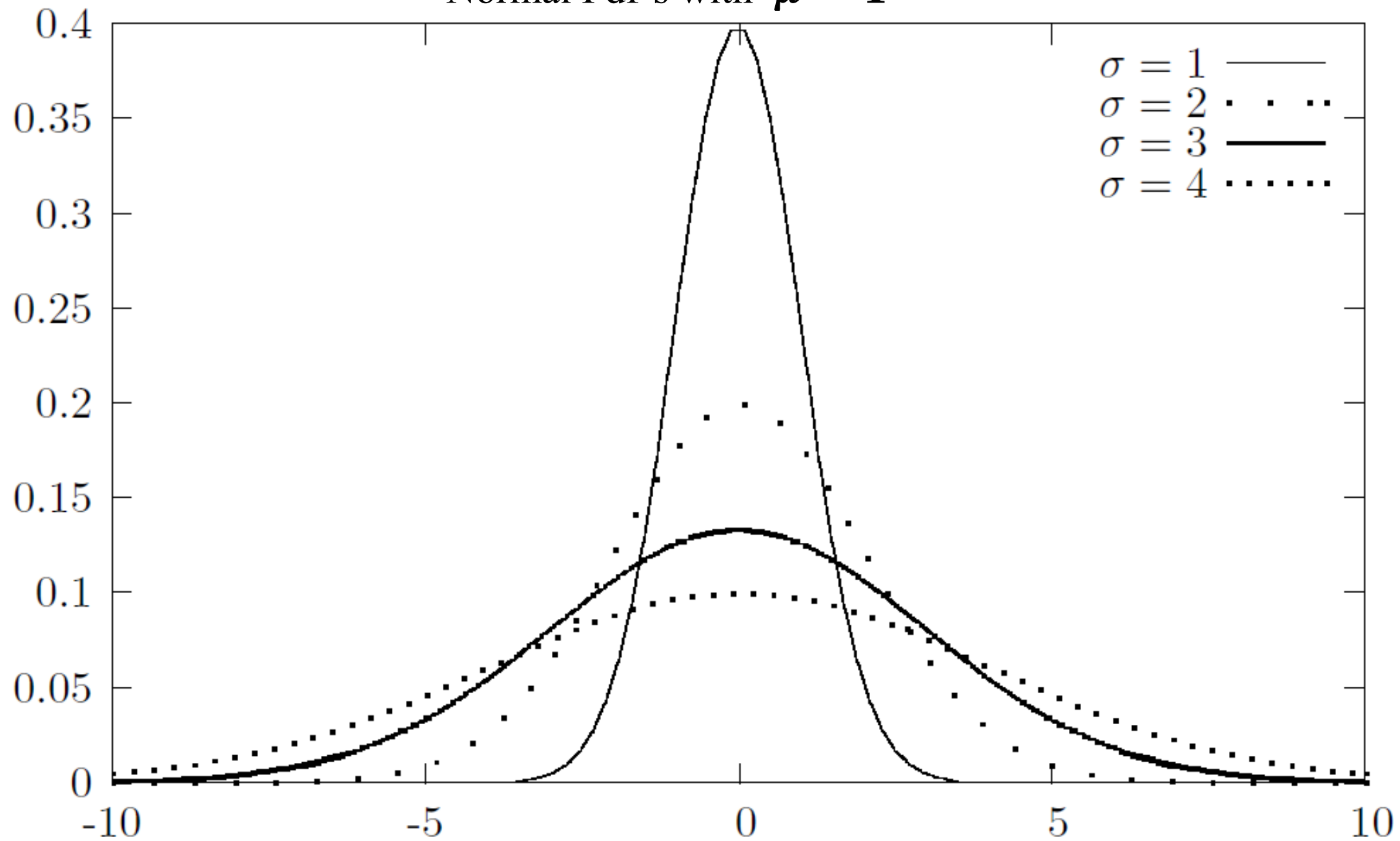
$$E(e^{tX}) = e^{(t\mu + t^2\sigma^2/2)}$$

Recurrence relation

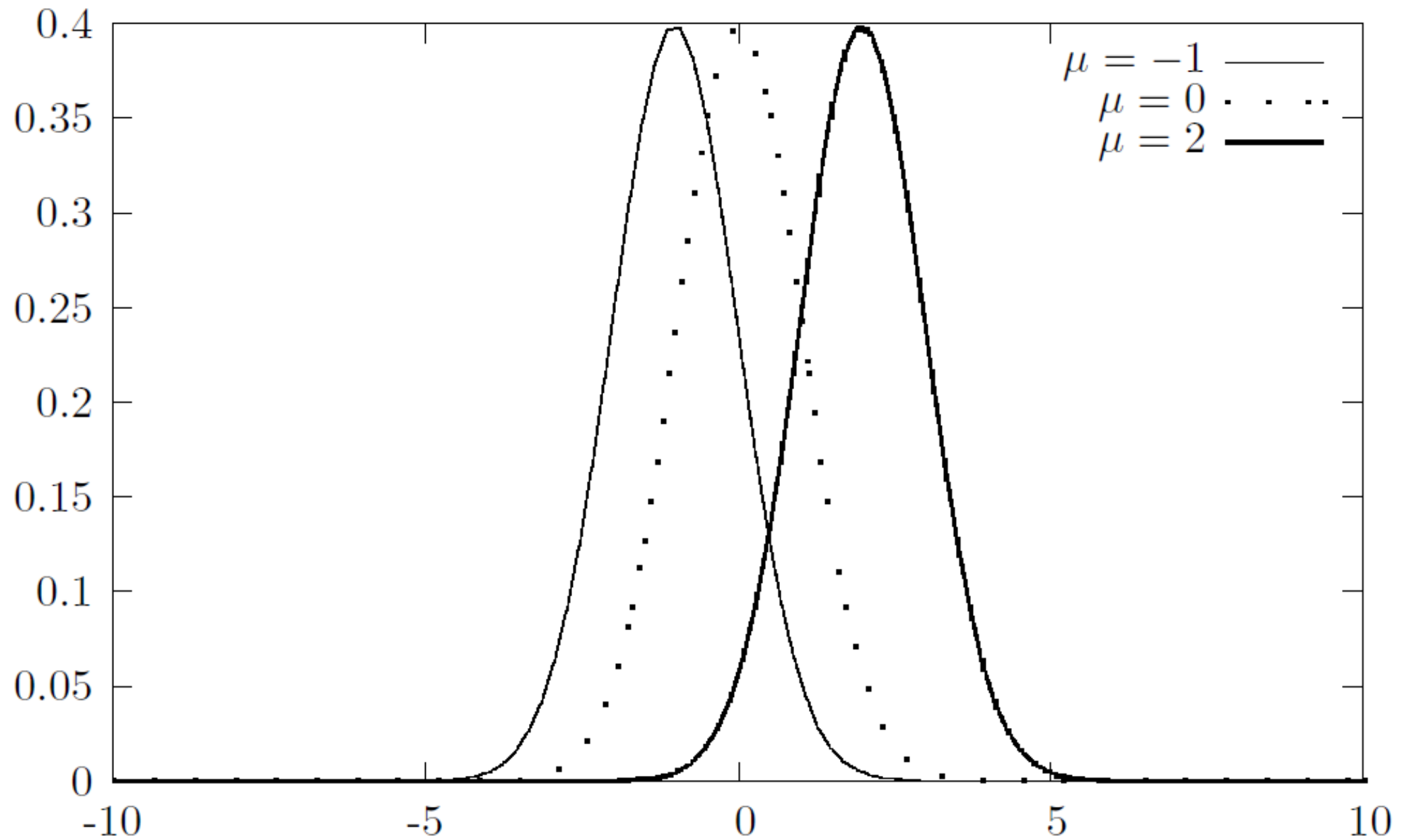
$$E(X^k) = (k-1)\sigma^2 E(X^{k-2}) + \mu E(X^{k-1})$$

$$\text{for } k = 3, 4, \dots$$

Normal Pdf's with $\mu = 1$



Normal Pdf's with $\sigma = 1$



Example

If X is a normally distributed r.v. with the mean of 12 and S.D. is 4.

Find out the probability of the following:

(i) $P(X \geq 20)$ (ii) $P(0 \leq X \leq 12)$

Solution:

Given that, $\mu = 12, \quad \sigma = 4, \quad i.e., \quad N(12, 16)$

(i) $P(X \geq 20)$

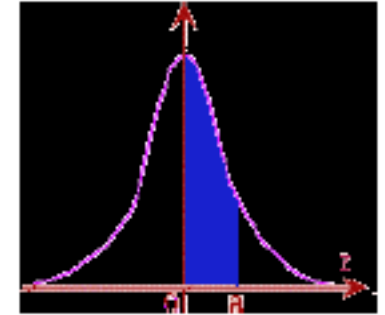
When, $X = 20, \quad Z = \frac{X - \mu}{\sigma} = \frac{20 - 12}{4} = 2$

$$\begin{aligned} \therefore P(X \geq 20) &= P(Z \geq 2) = 1 - P(Z \leq 2) \\ &= 1 - P(-\infty < Z \leq 2) \\ &= 1 - 0.9772 \end{aligned}$$

$$P(X \geq 20) = 0.0228$$

Or else, (i) $P(X \geq 20)$

$$\text{When, } X = 20, \quad Z = \frac{X - \mu}{\sigma} = \frac{20 - 12}{4} = 2$$



$$\begin{aligned} \therefore P(X \geq 20) &= P(Z \geq 2) &= 1 - P(Z \leq 2) \\ &= 1 - P(-\infty < Z \leq 2) \\ &= 1 - [P(-\infty < Z \leq 0) + P(0 \leq Z \leq 2)] \\ &= 1 - [0.5 + P(0 \leq Z \leq 2)] \\ &= 0.5 - P(0 \leq Z \leq 2) \\ &= 0.5 - 0.4772 \end{aligned}$$

$$P(X \geq 20) = 0.0228$$

$$(ii) \quad P(0 \leq X \leq 12) = P\left(\frac{0-12}{4} \leq \frac{X-\mu}{\sigma} \leq \frac{12-12}{4}\right)$$

$$= P(-3 \leq Z \leq 0)$$

$$= \Phi(0) - \Phi(-3)$$

$$P(0 \leq X \leq 12) = 0.5 - 0.0013 = 0.4986$$

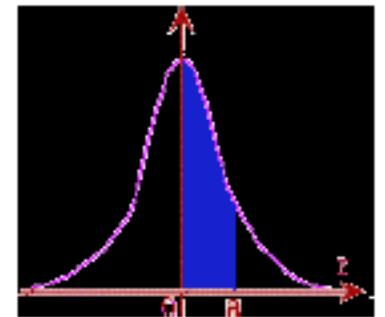
Or else,

$$P(0 \leq X \leq 12) = P\left(\frac{0-12}{4} \leq \frac{X-\mu}{\sigma} \leq \frac{12-12}{4}\right)$$

$$= P(-3 \leq Z \leq 0)$$

$$= P(0 \leq Z \leq 3)$$

$$P(0 \leq X \leq 12) = 0.4986$$



Example

If X is a normally distributed r.v. with mean 30 and S.D. 5.

Find the probabilities that:

(i) $26 \leq X \leq 40$ (ii) $X \geq 45$ (iii) $|X - 30| > 5$

Solution:

Given, $\mu = 30$, $\sigma = 5$, *i.e.*, $N(30, 25)$

$$(i) \ P(26 \leq X \leq 40) = P\left(\frac{26-30}{5} \leq \frac{X-\mu}{\sigma} \leq \frac{40-30}{5}\right)$$

$$= P(-0.8 \leq Z \leq 2)$$

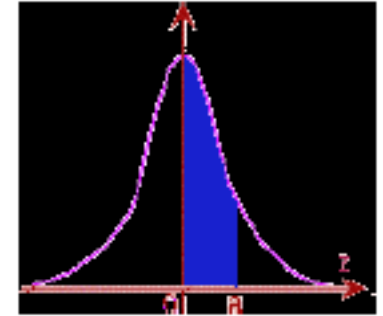
$$= \Phi(2) - \Phi(-0.8)$$

$$= 0.9772 - 0.2119$$

$$P(26 \leq X \leq 40) = 0.7653$$

Given, $\mu = 30$, $\sigma = 5$, *i.e.*, $N(30, 25)$

$$\begin{aligned} \text{(i) } P(26 \leq X \leq 40) &= P\left(\frac{26-30}{5} \leq \frac{X-\mu}{\sigma} \leq \frac{40-30}{5}\right) \\ &= P(-0.8 \leq Z \leq 2) \\ &= P(-0.8 \leq Z \leq 0) + P(0 \leq Z \leq 2) \\ &= P(0 \leq Z \leq 0.8) + P(0 \leq Z \leq 2) \\ &= 0.2881 + 0.4772 \end{aligned}$$



$$P(26 \leq X \leq 40) = 0.7653$$

$$\begin{aligned}
\text{(ii)} \quad P(X \geq 45) &= P\left(\frac{X-\mu}{\sigma} \geq \frac{45-30}{5}\right) \\
&= P(Z \geq 3) \\
&= 1 - P(Z \leq 3) \\
&= 1 - P(-\infty < Z \leq 3) \\
&= 1 - [0.5 + P(0 \leq Z \leq 3)] \\
&= 1 - 0.9987 \\
P(X \geq 45) &= 0.0013
\end{aligned}$$

$$(ii) \quad P(X \geq 45) = P\left(\frac{X-\mu}{\sigma} \geq \frac{45-30}{5}\right)$$

$$= P(Z \geq 3)$$

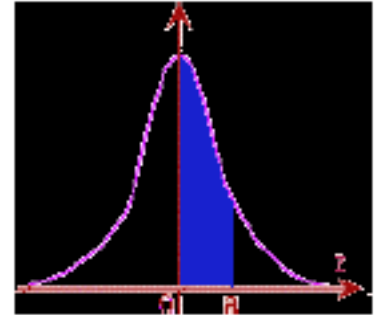
$$= 1 - P(Z \leq 3)$$

$$= 1 - [P(-\infty < Z \leq 0) + P(0 \leq Z \leq 3)]$$

$$= 1 - [0.5 + P(0 \leq Z \leq 3)]$$

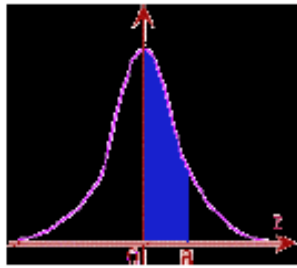
$$= 0.5 - 0.49865$$

$$P(X \geq 45) = 0.0013$$



$$\begin{aligned}
(iii) \quad P(|X - 30| > 5) &= 1 - P(|X - 30| \leq 5) \\
P(|X - 30| \leq 5) &= P(-5 \leq (X - 30) \leq 5) \\
&= P(-5 + 30 \leq X \leq 5 + 30) \\
&= P(25 \leq X \leq 35) \\
&= P\left(\frac{25-30}{5} \leq \frac{X-\mu}{\sigma} \leq \frac{35-30}{5}\right) \\
&= P(-1 \leq Z \leq 1) \\
&= \Phi(1) - \Phi(-1) \\
&= P(Z \leq 1) - P(Z \leq -1) \\
&= 0.8413 - 0.1587 \\
P(|X - 30| \leq 5) &= 0.6826 \\
P(|X - 30| > 5) &= 1 - P(|X - 30| \leq 5) \\
&= 1 - 0.6826 = 0.3174
\end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad P(|X - 30| > 5) &= 1 - P(|X - 30| \leq 5) \\
 P(|X - 30| \leq 5) &= P(-5 \leq (X - 30) \leq 5) \\
 &= P(-5 + 30 \leq X \leq 5 + 30) \\
 &= P(25 \leq X \leq 35) \\
 &= P\left(\frac{25-30}{5} \leq \frac{X-\mu}{\sigma} \leq \frac{35-30}{5}\right) \\
 &= P(-1 \leq Z \leq 1) \\
 &= P(-1 \leq Z \leq 0) + P(0 \leq Z \leq 1) \\
 &= P(0 \leq Z \leq 1) + P(0 \leq Z \leq 1) \\
 &= 2 P(0 \leq Z \leq 1) \\
 &= 2 \times 0.3413 \\
 P(|X - 30| \leq 5) &= 0.6826 \\
 P(|X - 30| > 5) &= 1 - P(|X - 30| \leq 5) \\
 &= 1 - 0.6826 = 0.3174
 \end{aligned}$$



For $\alpha \in (0, 1)$, let z_α be such that

$$P(Z \geq z_\alpha) = 1 - \Phi(z_\alpha) = \alpha$$

The value of z_α can, for any α , be obtained from Table

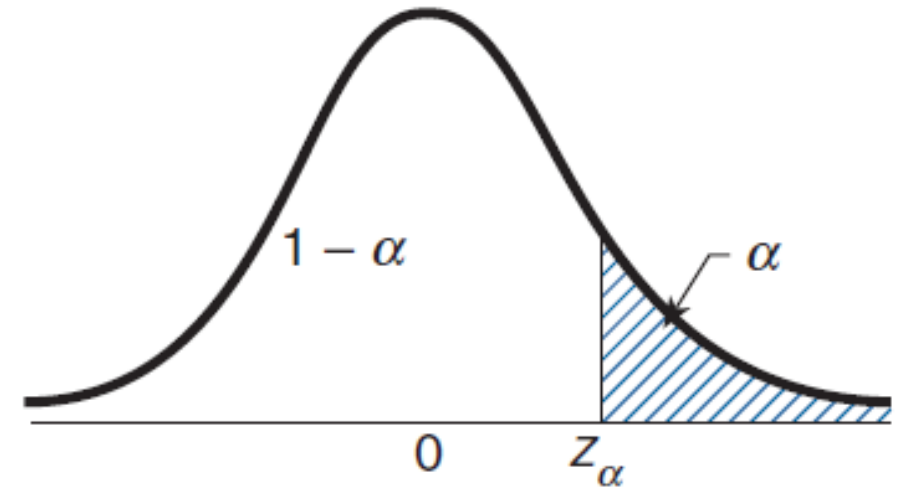
$$1 - \Phi(1.645) = 0.05$$

$$1 - \Phi(1.96) = 0.025$$

$$1 - \Phi(2.33) = 0.01$$

it follows that

$$z_{0.05} = 1.645, \quad z_{0.025} = 1.96, \quad z_{0.01} = 2.33$$



Since

$$P\{Z < z_\alpha\} = 1 - \alpha$$

it follows that $100(1 - \alpha)$ percent of the time a standard normal random variable will be less than z_α .

As a result, we call z_α the $100(1 - \alpha)$ percentile of the standard normal distribution.

Example

The mean yield for one-acre plot is 662 kilos with a s.d. 32 kilos. Assuming normal distribution, how many one-acre plots in a batch of 1,000 plots would you expect to have yield (i) over 700 kilos, (ii) below 650 kilos, and (iii) what is the lowest yield of the best 100 plots?

Solution:

If the random variable X denotes the yield (in kilos) for one-acre per plot,

Then $X \sim N(\mu, \sigma^2)$, where $\mu = 662$ and $\sigma = 32$.

(i) The probability that a plot has a yield over 700 kilos is given by

$$\begin{aligned} P(X > 700) &= P\left(\frac{X-\mu}{\sigma} > \frac{700-662}{32}\right) \\ &= P(Z > 1.19) \\ &= 1 - P(Z \leq 1.19) \\ &= 1 - P(-\infty < Z \leq 1.19) \\ &= 1 - [P(-\infty < Z \leq 0) + P(0 \leq Z \leq 1.19)] \\ &= 1 - [0.5 + P(0 \leq Z \leq 1.19)] \\ &= 0.5 - P(0 \leq Z \leq 1.19) \\ &= 0.5 - 0.3830 \end{aligned}$$

$$P(X > 700) = 0.117$$

$$\begin{aligned}
 P(X > 700) &= P\left(\frac{X-\mu}{\sigma} > \frac{700-662}{32}\right) \\
 &= P(Z > 1.19) \\
 &= 1 - P(Z \leq 1.19) \\
 &= 1 - P(-\infty < Z \leq 1.19) \\
 &= 1 - 0.8830 \\
 &= 0.117
 \end{aligned}$$

Hence in a batch of 1000 plots has a yield over 700 kilos is given by

$$1000 \times P(X > 700) = 117$$

(ii) Requires number of plots with yield below 650 kilos is given by:

$$\begin{aligned} P(X < 650) &= P(X < 650) = P\left(\frac{X-\mu}{\sigma} < \frac{650-662}{32}\right) \\ &= P(Z < -0.38) \\ &= P(Z > 0.38) \\ &= 1 - P(-\infty < Z \leq 0.38) \\ &= 1 - [P(-\infty < Z \leq 0) + P(0 \leq Z \leq 0.38)] \\ &= 1 - [0.5 + P(0 \leq Z \leq 0.38)] \\ &= 0.5 - P(0 \leq Z \leq 0.38) \\ &= 0.5 - 0.148 \end{aligned}$$

$$P(X < 650) = 0.352$$

Hence in a batch of 1000 plots has a yield over 650 kilos is given by

$$1000 \times P(X < 650) = 352$$

(iii) The lowest yield, say x_1 of best 100 plots is given by :

$$P(X > x_1) = \frac{100}{1000} = 0.1$$

$$\text{When } X = x_1, Z = \frac{x_1 - \mu}{\sigma} = \frac{x_1 - 662}{32} = z_1$$

Such that ,

$$P(Z > z_1) = 0.1$$

$$1 - [P(-\infty < Z \leq 0) + P(0 \leq Z \leq z_1)] = 0.1$$

$$0.5 + P(0 \leq Z \leq z_1) = 0.1$$

$$P(0 \leq Z \leq z_1) = 0.4$$

$$\Rightarrow z_1 = 1.28 \text{ (approx)}$$

$$\therefore x_1 = 662 + 32 z_1 = 662 + 32 \times 1.28 = 702.96$$

Hence the best 100 plots have yield over 702.96 kilos.

Example

The marks obtained by a number of students for a certain subject are assumed to be approximately normally distributed with mean value 65 and with a standard deviation of 5. If 3 students are taken at random from the set, what is the probability that exactly 2 of them will have marks over 70? [Ans: 0.0635]

Example

Data from the National Oceanic and Atmospheric Administration indicate that the yearly precipitation in a city is a normal random variable with a mean of 12.08 inches and a standard deviation of 3.1 inches.

Find the probability that the total precipitation during the next 2 years will exceed 25 inches.

Example

The effective life of dry battery cells is considered to be normally distributed with mean 20 hours and standard deviation 5 hours. In a consignment of 100 battery cells, how many cells are expected to have life,

- | | | |
|-------|-------------------------|--------------|
| (i) | less than 15 hours | [Ans: 15.88] |
| (ii) | between 8 and 16 hours | [Ans: 20.36] |
| (iii) | more than 15 hours | [Ans: 84.13] |
| (iv) | between 15 and 25 hours | [Ans: 68.26] |
| (v) | more than 30 hours. | [Ans: 2.27] |

Importance of N.D.

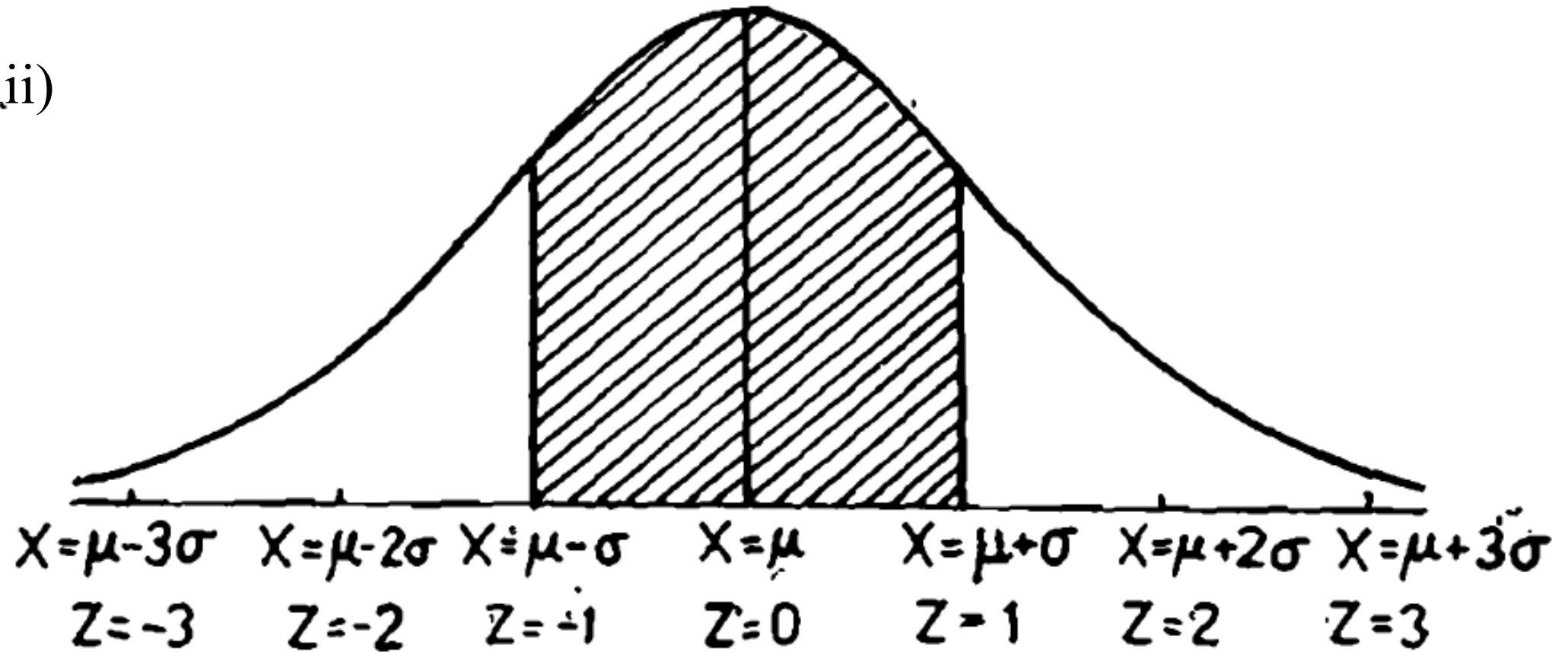
Normal distribution plays a very important role in statistical theory,

- (i) Most of the distributions occurring in practice, e.g., Binomial, Poisson, Hyper-geometric distributions, etc., can be approximated by normal distributions.

Moreover, many of the sampling distributions, e.g., Student's- t , Snedecor's F and Chi-square distributions, etc., tend to normality for large samples.

- (ii) Even if a variable is not normally distributed, it can be sometimes be brought to normal form by simple transformation of variable.

(iii)



This property is the basis of entire Large sample theory.

- (iv) The entire theory of small sample tests, t , F and χ^2 tests, etc., is based on the fundamental assumption that the parent populations from which the samples have been follow normal distribution.
- (v) Normal distribution finds large applications in Statistical quality control in industry for setting up control limits.